Automated Failure Attribution for LLM-Based Multi-Agent Systems

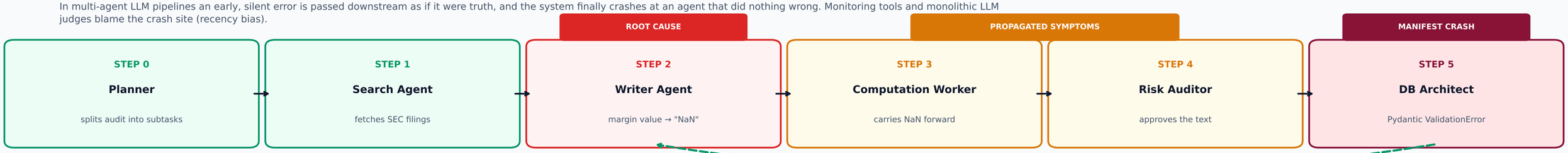
A Neuro-Symbolic Causal Graph Engine that finds the root cause — not the crash site

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Guide: Dr. D. Ushasree · Dept. of CSE, GRIET · Major Project Phase-I, 2026–27

1 THE PROBLEM: Symptom Misattribution

In multi-agent LLM pipelines an early, silent error is passed downstream as if it were truth, and the system finally crashes at an agent that did nothing wrong. Monitoring tools and monolithic LLM judges blame the crash site (recency bias).



✗ **Baselines blame Step 5: only 14.2% step-level accuracy (All-at-Once LLM judge)**

✓ **Our engine walks the dependency DAG backward and isolates Step 2 as the true origin**

2 THE SOLUTION: Neuro-Symbolic Causal Graph Engine

Neural (local LLM) · Symbolic (graph algorithms) · Engineering

1 Trace Segmentation & Token Budgeting

Parses AutoGen, LangGraph, CrewAI and MetaGPT logs into ExecutionSteps. 60% head / 30% tail retention keeps prompts inside a 4,096-token budget.

ENGINEERING

2 Cached Semantic Step Evaluation

Local Qwen2.5:7b (Ollama, JSON mode) scores each turn in isolation. SHA256 cache (memory + SQLite) removes redundant inference; MockJudge is the offline fallback.

NEURAL

3 Dependency DAG Synthesis

NetworkX DiGraph with sequential edges (confidence 0.85) and regex-extracted variable data-flow edges (confidence 0.95).

SYMBOLIC

4 Backward Causal Traversal

Computes the Transitive Ancestor Closure of the crash, then blames the earliest anomaly whose parents are all clean (Clean-Parent Invariant).

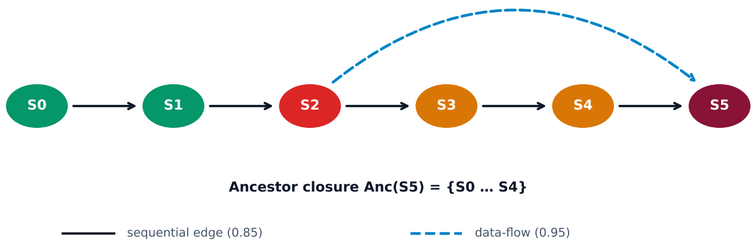
SYMBOLIC

5 2D Taxonomy & Visual Report

Causal Role × 25+ bug categories, shortest causal chain (Dijkstra), Streamlit dashboard and FastAPI REST endpoints for CI/CD.

ENGINEERING

3 ROOT-CAUSE ISOLATION



Earliest anomaly with clean parents → S2 = ROOT CAUSE

4 RESULTS · ICML 2025 Who&When (184 traces)



KEY OUTCOMES

85.3% Who accuracy	**78.3%** When accuracy
−91.2% inference latency	**100%** cache hit rate

27 / 27 automated tests pass

Tech stack: Python · Ollama (Qwen2.5:7b) · NetworkX · SQLite · FastAPI · Streamlit

Baseline figures as reported for the Who&When benchmark.